

SUSHMITHA SUDHARSAN

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SUMMARY

AI/ML Engineer with an MS in Data Analytics Engineering, specializing in production-grade MLOps and applied GenAI. Built end-to-end ML pipelines with automated CI/CD, retraining, and drift monitoring, sustaining sub-200ms latency and reducing data errors by 80%. Proficient in PyTorch, MLflow, Airflow, Docker, and GCP. First-author IEEE publication in applied deep learning.

EDUCATION

Northeastern University <i>Master of Science in Data Analytics Engineering, GPA: 3.92/4.0</i> Coursework: Data Mining, Generative AI, MLOps, Machine Learning, Deep Learning for AI, Data Visualization	Apr 2026 Boston, MA
SRM Institute of Science and Technology <i>Bachelor of Technology in Computer Science (AI & ML Specialization)</i> Coursework: Deep Learning, Artificial Intelligence, Robotics, Probability and Statistics	May 2024 Chennai, India

TECHNICAL SKILLS

Programming: Python, SQL, R, C, C++
Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM
GenAI & NLP: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Multi-Agent Systems, Natural Language Processing, Prompt Engineering
Data & Pipelines: Pandas, NumPy, Apache Airflow, ETL Pipelines, API Integration
MLOps & Deployment: MLflow, Docker, CI/CD, FastAPI, Streamlit, Model Monitoring, Prometheus, DVC, Great Expectations, PyTest
Cloud: Google Cloud Platform (GCP), Cloud Run, HuggingFace Spaces
Computer Vision: YOLOv8, Object Detection, Image Classification, OpenCV, CLIP Embeddings

PROJECTS

MLOps Financial Forecasting Platform [GitHub] <i>PyTorch, XGBoost, Airflow, MLflow, GCP</i>	Sep – Dec 2025
- Collaborated with a 6-member team to forecast how 100 companies' stocks would react to crisis scenarios like 2008 or COVID, automating PyTorch and XGBoost pipelines through Apache Airflow and validating 2.5M+ records with Great Expectations - Eliminated manual retraining bottlenecks by owning the CI/CD pipeline for automated retraining, GCP deployment, and PyTest unit testing, tracking 50+ MLflow experiments and monitoring drift with Prometheus/Grafana under 200ms p99 latency	
GenAI Sports Analysis Chatbot [GitHub] <i>LangChain, FAISS, GPT-4o, Gradio</i>	Mar 2026
- Replaced slow, manual stat lookups with a chatbot answering World Cup stats questions across 49,000+ matches and 47,000+ goals; designed the ReAct agent logic connecting GPT-4o, embeddings, FAISS, and 5 tools - Shipped the working chatbot in 48 hours with trustworthy, evidence-backed answers (confidence scores, stated limitations), plus a 5-factor match-outcome predictor via Gradio	
VisionScript: Multimodal Image Captioning [GitHub] <i>CLIP, BLIP, GPT-2, PyTorch</i>	Jan – Apr 2026
- Addressed the rigidity of standard image captions by building a tone-adjustable captioning pipeline (e.g., simplified, child-friendly) with a 4-member team; owned the CLIP-GPT-2 integration via a custom projection layer, fine-tuned on 1,000+ Flickr30k pairs - Owned decoding-strategy testing to reduce hallucinated captions and improve grounding, then deployed a live demo on HuggingFace Spaces	

EXPERIENCE

Data Analyst Intern <i>Whirl Data Inc.</i>	Jun 2023 – Sep 2023 Chennai, India
- Built a quality-check framework to process 10,000+ image frames through a data pipeline, reducing annotation errors by 80% - Optimized a computer vision model through systematic testing and evaluation, increasing accuracy from 73% to 88% - Applied and documented ML best practices with the team, reducing debugging time by 30% across the wider team	

ACHIEVEMENTS & PUBLICATIONS

- IEEE Publication (First Author):** Developed an RNN/LSTM model to track classroom engagement and fatigue by flagging when a student needs a break for peak focus, achieving 96% accuracy with 2–3 sec real-time latency across 30 participants
- Finalist, Data Mining Hackathon:** Led a team building an ML pipeline in 48 hours, achieving a 0.89 F1-score (2nd of 25 teams)
- Certifications:** Microsoft Azure Fundamentals (AZ-900) | Microsoft Azure AI Fundamentals (AI-901) | Microsoft Power BI | AWS Generative AI